AERIS: Boundary-Mapped Reliable Routing for Wireless Sensor Networks under Heterogeneous Channels

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***Abstract*—AERIS is a rule-based hierarchical routing protocol for wireless sensor networks under heterogeneous channels. It combines context-adaptive intra-cluster forwarding, Gateway-assisted CH-to-BS uplinks, and a sparse fallback backbone. An NS-3 rerun package evaluates AERIS against LEACH, PEGASIS, HEED, TEEN, CTP, and RPL-MRHOF across four environments and 50–1000 nodes ($n = 30$ per cell). The result is a deployment boundary rather than a global ranking: AERIS is rank-1 in 21 of 28 classical-baseline cells, but after adding CTP/RPL-MRHOF it remains strongest mainly in outdoor suburban cells. A 3360-run ablation and a 400-replicate mechanism study attribute the gain mainly to Gateway-assisted uplinks and expose the cost: early first-node death in harsh large-scale cells. AERIS is therefore a simple reliability-first option for selected harsh regimes, not a replacement for richer collection-tree or RPL-style stacks in all deployments.**

***Index Terms*—wireless sensor networks, reliable routing, cross-platform evaluation, deployment boundary**

I. Introduction

Wireless sensor network (WSN) routing is usually discussed in terms of energy, yet in many deployments link stability matters just as much [1]–[3]. This becomes clear when classical protocols are placed side by side: LEACH emphasizes clustered forwarding [4], PEGASIS uses chain-based aggregation [5], HEED relies on residual-energy-aware head election [6], and TEEN reports only when event thresholds are crossed [7]. Real links make the trade-off harder because low-power WSN channels are asymmetric, environment sensitive, and often unstable near connectivity thresholds [8]–[10].

This leads to a practical question: when do simple, auditable control rules improve on classical energy-first designs under harsh channels, and when are richer collection-tree or RPL-style stacks the better answer? AERIS addresses the first half of this question with fixed-rule control rather than learned routing. Its design combines Context-Adaptive Switching (CAS) for intra-cluster mode selection, Gateway reinforcement for CH-to-BS uplinks, and a sparse Skeleton fallback backbone.

We make three contributions. First, we present AERIS as a fixed-rule routing design whose online behavior can be traced directly to channel and topology inputs. Second, we evaluate AERIS with a five-protocol NS-3 rerun, an expanded seven-protocol CTP/RPL-MRHOF boundary sweep, and a stricter simulator matrix, keeping baseline ranking separate from stress evidence. Third, we use ablation, pooled trade-off statistics, and a mechanism study to show that the observed gain comes mainly from Gateway support and is paid for through earlier node death.

II. Related Work

Classical low-overhead WSN protocols still define the main engineering operating points. LEACH and HEED reduce long-range transmission burden through clustering, but their uplink quality is still sensitive to head placement and channel degradation [6], [11]. PEGASIS spreads load through chain forwarding but pays for that robustness with long routes and serialization [5]. TEEN reduces redundant transmissions under threshold-triggered sensing, but its reporting logic makes metric interpretation sensitive to how "expected" delivery is defined [7]. These protocols span the basic reliability-energy-latency trade-offs that any new design has to face.

Recent work has improved reliability through learning, hybrid control, context-aware routing, or deployment adaptation [12]–[18]. AERIS is adjacent to this line of work, but it keeps the runtime logic deliberately simple: fixed coefficients, explicit control paths, and no training-time dependence. Recent 2024–2025 surveys further show the same drift toward richer control planes, LLN management, and AI-driven routing [19]–[21]. Concrete routing designs continue to explore reinforcement learning, mobile-sink adaptation, hybrid clustering, and federated DDQN strategies [22]–[25].

Collection and RPL-style protocols are especially important for deployment realism. CTP builds a collection tree toward the sink using link/path quality [26], while RPL with MRHOF selects parents by minimizing path cost in a low-power and lossy-network DAG. Opportunistic variants such as ORPL/ORW go further by exploiting candidate forwarder sets and local link diversity. IEEE LCN has also carried adjacent WSN/IoT work, from SmartCoast to mobility-prediction routing, OSR, and decentralized broadcast trees [27]–[30]. This paper does not claim to replace that whole family. Instead, it asks where a simpler fixed-rule WSN design still buys reliability, and where the stronger collection-tree family should remain the default.

III. Method

AERIS combines CAS for intra-cluster transmission mode selection, Gateway reinforcement for cluster-head to base-

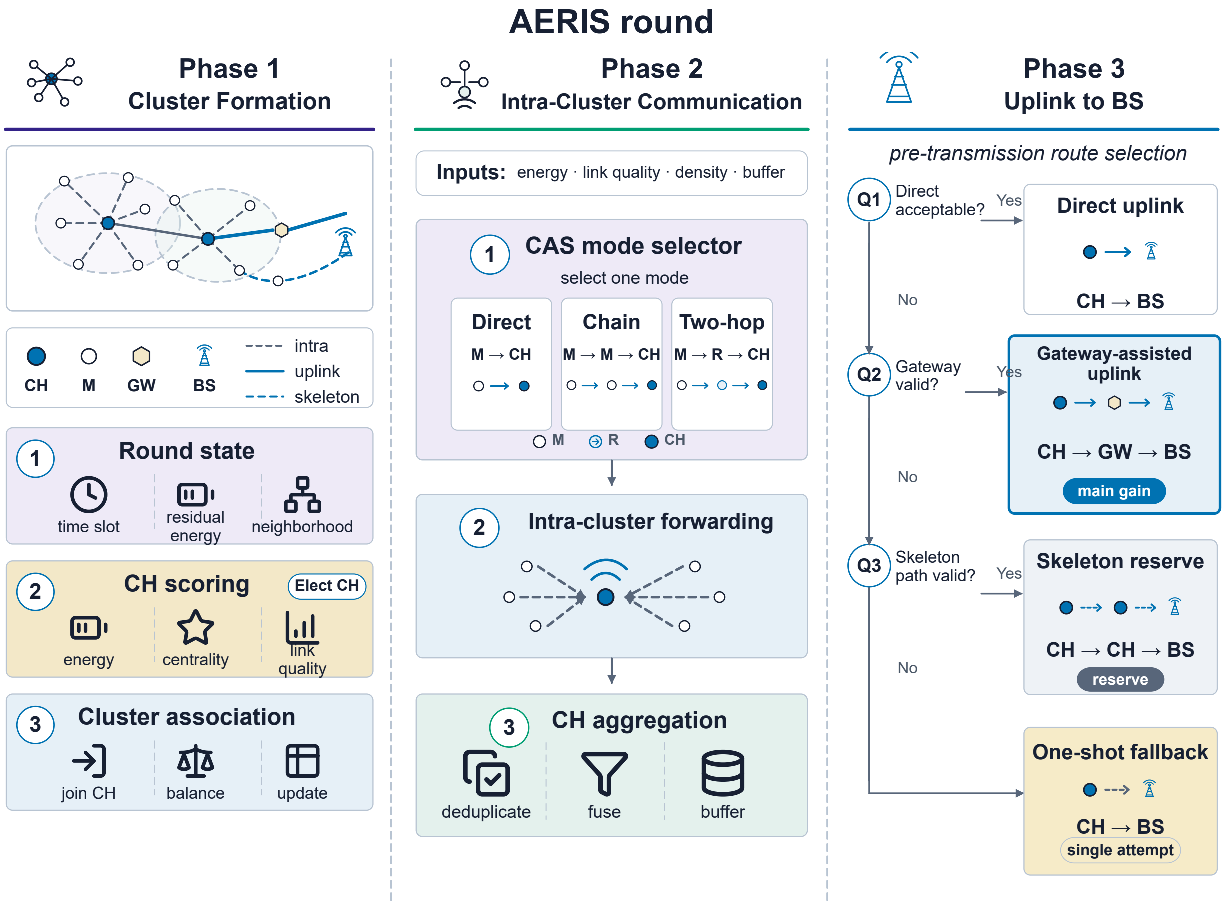

Fig. 1. AERIS round framework. Phase 1 combines an abstract topology inset with a compact legend that maps the role symbols M, R, CH, GW, and BS, together with intra-cluster, uplink, and skeleton links, and three process cards for round-state sensing, CH scoring, and cluster association. Phase 2 contains the CAS selector with direct, chain, and two-hop modes, the sparse intra-cluster forwarding sketch, and CH-level packet aggregation. Phase 3 places a pre-transmission decision spine beside four route outcomes: direct uplink, Gateway-assisted uplink as the main gain path, Skeleton reserve, and one-shot fallback.

station uplinks, and a sparse Skeleton CH relay backbone as a reserve fallback. The goal is to add reliability support only where direct forwarding becomes unreliable, while keeping online logic deterministic and auditable. Figure 1 summarizes the framework in the same order: cluster formation, intra-cluster communication, and pre-transmission uplink selection to the BS.

Gateway candidates are ranked by a fixed linear score,

$$\mathrm{Score}_i = \alpha \hat{E}_i + \beta \hat{C}_i + \gamma \hat{L}_i + \delta \hat{D}_i, \tag{1}$$

where $\hat{E}_i$, $\hat{C}_i$, $\hat{L}_i$, and $\hat{D}_i$ denote normalized residual energy, centrality, link quality, and CH-to-BS distance. Centrality is a distance-aware neighborhood surrogate computed from local geometry, link quality is the current channel-estimated success probability for the candidate hop, and all scalar features are normalized to $[0, 1]$ before scoring. CAS uses another fixed linear rule over {direct, chain, two-hop}; the shared coefficients are fixed before all runs and are not tuned per environment:

$$m^{\star} = \arg \max_{m \in \{\mathrm{direct},\mathrm{chain},\mathrm{two\text{-}hop}\}} \left(a_m \hat{q} + b_m \hat{e} + c_m \hat{u} \right), \tag{2}$$

where $\hat{q}$, $\hat{e}$, and $\hat{u}$ are normalized local channel quality, residual energy, and utilization terms, and the coefficients $\{a_m, b_m, c_m\}$ are fixed for each candidate mode. Skeleton forwarding is built as a sparse CH relay backbone after clustering and refreshed with the CH set; in this publication configuration it acts as a reserve fallback rather than an actively exercised mechanism.

The forwarding logic is cascaded rather than multipath. A cluster head first checks direct uplink quality, then tries Gateway support if available, then Skeleton support if available, and finally makes one last direct attempt. In the strict-physics regime, this "best-effort direct fallback" is a single direct transmission attempt without protocol-level retransmission.

In practice, CAS decides whether a sensor-to-CH transfer stays direct, switches to a short chain, or uses a two-hop relay; Gateway support then targets the more fragile CH-to-BS segment; Skeleton remains a reserve fallback if Gateway support is unavailable. AERIS is therefore an engineering design rather than a formal routing theorem: the same inputs trigger the same routing decisions, and the design is judged by how those decisions change delivery and cost in deployment.

The module order is deliberate: direct transmission is checked first because it is cheapest when the channel supports it; Gateway support comes next because the CH-to-BS segment is usually the first harsh-cell bottleneck; Skeleton is placed last because it preserves a fallback path rather than replacing the main uplink rule. The design is aimed at deployments where channel quality varies across the field and modest runtime complexity is acceptable if it stabilizes delivery, not at clean office-like regimes or systems that can already afford richer LLN control.

IV. Experiments

A. Experimental Setup

The primary metric is packet delivery ratio over expected application-layer transmissions,

$$\mathrm{PDR}_{\mathrm{expected}} = \frac{N_{\mathrm{delivered}}}{N_{\mathrm{expected}}}, \tag{3}$$

where N_{expected} counts one packet per alive source node per round even if a protocol suppresses actual transmission. This denominator is kept fixed across protocols so that the reliability comparison is not quietly changed by protocol-specific reporting logic.

We use complementary settings. A five-protocol NS-3 rerun compares AERIS with LEACH, PEGASIS, HEED, and TEEN across four environments and 100, 500, and 1000 nodes ($n = 30$ per cell). An expanded split-machine NS-3 boundary sweep adds CTP and RPL-MRHOF over 50–1000 nodes. In this standalone harness, CTP and RPL-MRHOF are simplified collection-style baselines, not complete LLN stacks: CTP uses progress-constrained link-cost forwarding, while RPL-MRHOF uses accumulated path cost for parent/path selection. A 3360-run NS-3 ablation compares full AERIS with no-Gateway, no-CAS, and no-CH-score variants. A strict-physics Python matrix adds collision and relay stress; LEACH, HEED, and TEEN are minimally relay-enabled there for multi-hop comparability. The fixed 100-node block reports PDR, total energy, lifetime, first-node death, and hop count, while a 400-replicate mechanism matrix explains the active AERIS mechanisms.

For the NS-3 sweeps, nodes are uniformly placed in a 200 m $\times$ 200 m field with the sink at the top-center boundary; runs use 300 rounds, 512-byte packets, 2.0 J initial energy, 10 dBm transmit power, and seeds 42001–42030. The channel combines log-distance loss, shadowing, fading, receiver sensitivity, and packet-error probability. The path-loss/shadowing pairs are office (2.0, 4.5 dB), factory (2.7, 8.5 dB), suburban (2.8, 7.5 dB), and urban (3.4, 12.0 dB). The harness remains routing-level rather than a full 802.15.4 MAC stack.

This role split is deliberate. The five-protocol NS-3 rerun anchors the classical-baseline comparison; the seven-protocol sweep maps the CTP/RPL-MRHOF boundary; the ablation attributes the gain; the strict-physics layer is stress evidence rather than a second canonical ranking. Lifetime, FND, hops, and energy expose load concentration. In Fig. 4, dots use Welch-type cell tests with Hedges' g and Holm correction; in Fig. 2, gaps below 0.1 percentage point are treated as near-ties.

TABLE I
Experimental Settings Used in the Paper.

Layer	Platform	Protocols	Cells	Role
Classical NS-3	NS-3	5 protocols	4×3	anchor
Boundary sweep	NS-3	7 protocols	4×7	boundary
Module ablation	NS-3	4 AERIS variants	4×7	attribution
Strict	Python	adapted 5-protocol	4×6	stress
Mechanism	Python	AERIS-only grid	4×3	attribution
Frozen block	pooled stats	5-protocol summary	100-node only	trade-off

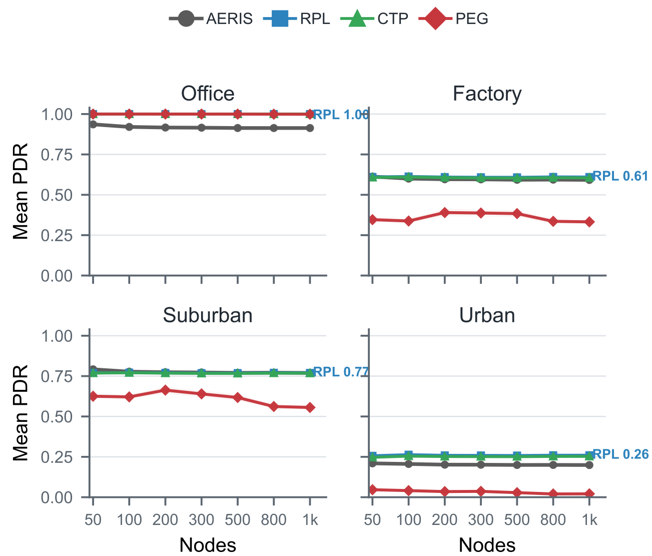

Fig. 2. Expanded NS-3 boundary sweep across four environments and 50–1000 nodes ($n = 30$ per cell). Curves show mean PDR for AERIS and the strongest collection/classical competitors; shaded bands denote 95% confidence intervals.

B. Expanded Boundary Sweep

Figure 2 gives the main boundary result from the seven-protocol split-machine NS-3 sweep. If the baseline family is restricted to classical WSN protocols, AERIS is rank-1 in 21 of 28 environment-node cells and top-2 in all 28. This supports the classical-baseline claim: AERIS is a strong reliability-first design relative to LEACH, PEGASIS, HEED, and TEEN outside the benign office regime.

The expanded sweep also shows why that claim must be bounded. Once CTP and RPL-MRHOF are included, AERIS remains strongest mainly in outdoor suburban cells: it wins 6 of 7 scales and is essentially tied at 1000 nodes, where RPL-MRHOF leads by only 0.0001 PDR points in the split-machine sweep. Such cells should be read as near-ties rather than practical superiority claims. In contrast, CTP leads the office cells, and RPL-MRHOF leads most factory and urban cells. AERIS is slightly ahead at factory-50, but falls 1.3–1.8 percentage points behind RPL-MRHOF at larger factory scales; in urban, the average gap to the winner is 5.69 points.

This boundary is the main practical outcome of the expanded sweep. AERIS solves a weakness of the classical clustering/chain baselines by protecting fragile uplinks with

Gateway support, but collection-tree and RPL-style baselines already solve part of that path-selection problem with richer parent choice. The strongest defensible claim is therefore not that AERIS is globally best, but that its rule-based control path delivers a large gain over classical WSN baselines and remains competitive or best in selected harsh-channel regimes, especially suburban cells.

C. Classical Audit and Stress Evidence

The five-protocol NS-3 audit remains useful as the classical-baseline anchor. In that subset, PEGASIS is best in indoor office, while AERIS leads in factory, suburban, and urban conditions. At 1000 nodes, AERIS reaches 0.593 in indoor factory, 0.770 in outdoor suburban, and 0.200 in outdoor urban, compared with 0.324, 0.549, and 0.019 for PEGASIS. This is why the expanded result should be read as a boundary refinement rather than a reversal of the classical-baseline claim.

The strict-physics matrix asks whether the classical-baseline ordering survives once collision and relay stress are made explicit in the adapted layer. Figure 3 shows that it does. PEGASIS remains strongest in indoor office (0.991 at 100 nodes and 0.988 at 1000), whereas AERIS stays rank-1 in indoor factory, outdoor suburban, and outdoor urban. At 1000 nodes, AERIS remains well above PEGASIS in indoor factory (0.728 vs. 0.300) and outdoor suburban (0.728 vs. 0.473), and it still leads in outdoor urban (0.136 vs. 0.099).

This matrix is not the main fairness anchor because LEACH, HEED, and TEEN are minimally augmented with a simple relay layer for multi-hop comparability in this setting. We therefore do not interpret Fig. 3 as a second canonical protocol ranking. Its role is narrower: to show that AERIS's harsh-channel advantage is not erased by explicit collision-and-relay stress even though absolute PDR falls for every protocol in the harder scales.

D. Ablation and Deployment Boundary

Figure 4 gives the expanded NS-3 module attribution test. It reruns full AERIS and three ablated variants over the four-environment, seven-scale matrix used in the boundary sweep, for 3360 experiments in total. The result is sharper than the earlier fixed 100-node ablation: Gateway is the main gain carrier in factory and suburban cells. Removing Gateway costs 5.82 percentage points on average in factory and 7.58 points in suburban, with all seven node scales significant after Holm correction in both environments. It is essentially neutral in office and weak in urban, which is consistent with the boundary result: office does not need extra uplink protection, while urban links can become too poor for a nearby Gateway to rescue reliably.

CAS behaves differently. Removing CAS is beneficial in office and suburban cells, but hurts urban delivery by 0.91 points on average, with five of seven urban scales significant. This means CAS is not a universal gain module; it is useful mainly when a short local detour remains viable under harsh links. Removing the CH scoring component is near-neutral in all environments. This does not make the score irrelevant to the design, but it does show that the measured delivery gain in this publication configuration should not be attributed to head scoring.

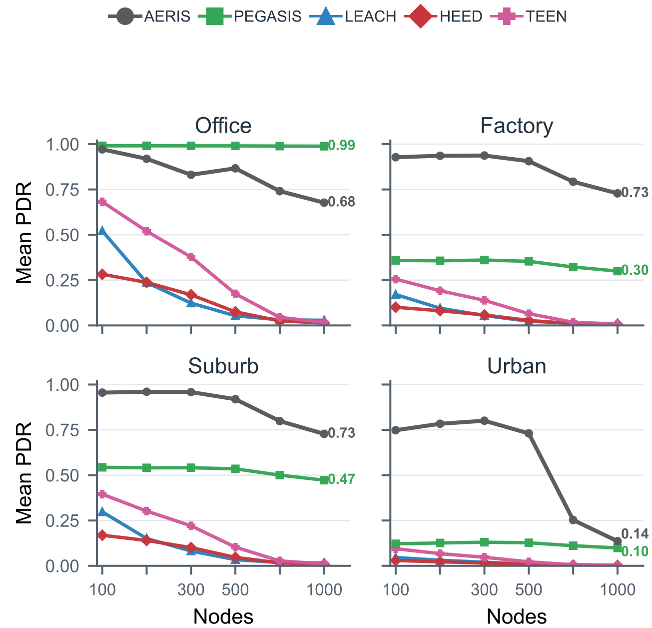

Fig. 3. Strict-physics Python adapted stress test across four environments and 100–1000 nodes ($n = 3200$ per cell). LEACH, HEED, and TEEN are minimally relay-enabled here; this layer tests collision-and-relay stress and is not canonical ranking evidence.

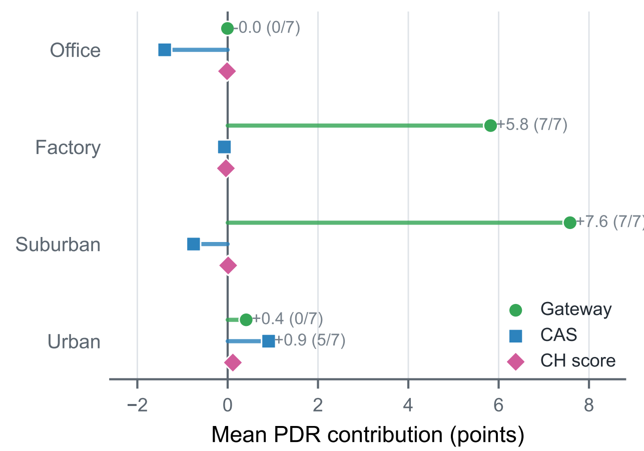

Fig. 4. Expanded NS-3 AERIS ablation summary across four environments and 50–1000 nodes ($n = 30$ per cell). Points show mean PDR contribution of each module; labels give contribution and Holm-significant node-count cells.

E. Mechanism and Deployment Trade-off

The 400-replicate mechanism matrix is summarized in Fig. 5. Gateway uplink PDR stays near 1.0 in office, factory, and suburban cells, but drops sharply at urban-1000, exactly where end-to-end PDR collapses. A targeted follow-up rerun of the three harsh 1000-node cells on an independent machine reproduced the same bottleneck pattern to the reported

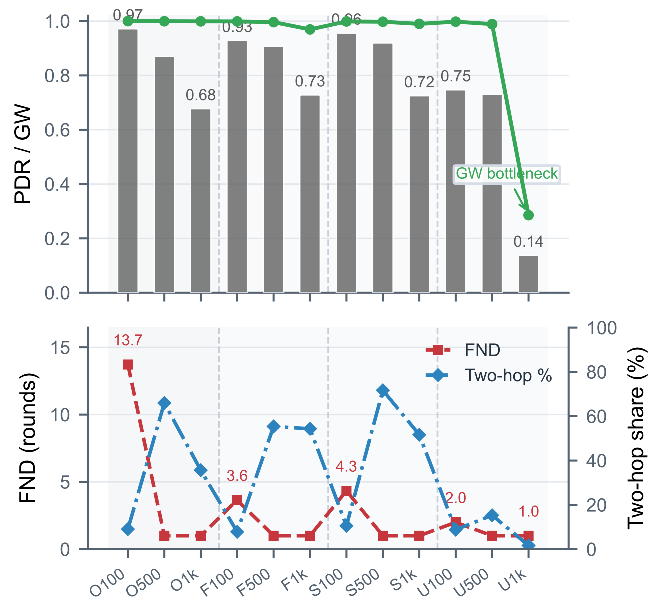

Fig. 5. AERIS mechanism matrix from 400 replicates across 12 environment-node cells. The top panel shows end-to-end and Gateway-uplink PDR; the lower panel shows first-node death (FND) and CAS two-hop share. Early FND exposes the reliability-lifetime cost of concentrated relay roles.

TABLE II

FIXED 100-NODE BLOCK. BOLD MARKS THE PROPOSED METHOD. AERIS HAS THE HIGHEST POOLED PDR AND LOWEST TOTAL ENERGY, BUT ALSO THE SHORTEST LIFETIME AND EARLY FND; FND=0.0 DENOTES NO OBSERVED FIRST DEATH BEFORE THE HORIZON.

Method	PDR↑	Energy↓	Lifetime↑	FND↑	Hops↓
LEACH	0.260	165.4	300.0	0.0	1.71
HEED	0.414	165.2	300.0	11.4	2.00
TEEN	0.432	159.4	299.0	0.0	1.28
PEGASIS	0.373	137.6	300.0	70.7	32.37
AERIS	**0.674**	**131.1**	**215.5**	**15.8**	**1.98**

precision. The visual takeaway is the same as the ablation: Gateway carries the gain, CAS is conditional, and Skeleton remains dormant because Gateway support resolves most eligible fallback cases in this configuration.

The cost is just as clear. Table II shows that AERIS has the lowest total energy (131.1 J) but the shortest lifetime (215.5 rounds). Fig. 5 localizes the cost to early node death and shrinking two-hop support in harsh cells. Better delivery is bought with concentrated relay burden; this is the central engineering trade-off, not a secondary artifact.

Figures 2–4 and Table II suggest a concrete deployment rule. Prefer PEGASIS-like routing when lifetime dominates in benign cells; prefer CTP/RPL-style routing when richer collection-tree control is acceptable in office, factory, or urban regimes; choose AERIS when delivery under heterogeneous harsh links is the priority and a simple auditable WSN controller is required, especially in suburban-like cells. The office regime, longevity-first deployments, and mature LLN stacks are the clearest negative cases.

V. LIMITATIONS

The main message is simple. AERIS is strongest as a simple reliability-first alternative to classical WSN baselines, especially in suburban harsh-channel cells; it is not the default for benign office-like regimes, lifetime-dominated deployments, or systems where collection-tree/RPL-style routing is available. Its gain should be attributed mainly to Gateway-assisted uplinks, not to all modules equally.

Several limits remain. The expanded NS-3 sweep includes CTP and RPL-MRHOF collection baselines, but not opportunistic routing, ORPL/ORW-style candidate forwarding, coding-based reliability, channel hopping, or standards-compliant RPL stacks. The strict-physics layer is deployment-stress evidence rather than external ground truth, and its MAC abstraction omits full 802.15.4/TSCH/CSMA-CA behavior. Recent LLN work likewise emphasizes scalable path management and control-plane flexibility [19], [20]. Because TEEN is event-triggered, $\mathrm{PDR}_{\mathrm{expected}}$ should be read with energy, lifetime, FND, and hops; actual-transmission PDR or event-detection success would be needed for a TEEN-centered study. The current study is also limited to static topologies, the tested environment taxonomy, and artifacts that are not yet public.

The provenance chain separates native-harness evidence from stress evidence: the five-protocol NS-3 rerun anchors the classical comparison, the seven-protocol split-machine sweep maps the CTP/RPL-MRHOF boundary, the strict-physics layer stress-tests that story, and the mechanism grid attributes the gain to Gateway support and selected two-hop use. A separate rerun of the three harsh 1000-node mechanism cells reproduced the key bottleneck means to the reported precision.

VI. CONCLUSION

This paper revisited AERIS as a reliable WSN routing design with a five-protocol NS-3 rerun, an expanded seven-protocol CTP/RPL-MRHOF boundary sweep, a strict-physics simulator matrix, and mechanism evidence. The claim is deliberately bounded: AERIS is a strong reliability-first alternative to classical WSN baselines, with gains carried mainly by Gateway support, but richer collection-tree routing narrows or removes that advantage in office, factory, and urban regimes. It also buys delivery by concentrating forwarding work, shortening lifetime and advancing first-node death. The practical contribution is therefore a visible deployment boundary rather than a global leaderboard.

Future work should compare against full RPL/ORPL/ORW-style stacks, add more realistic MAC behavior, and test whether Gateway-style reinforcement can be combined with collection-tree parent selection without reproducing the early-node-death cost.

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